Assignment 1

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1 Chapter 1

P6

a. $d_{\text{prop}} = \frac{m}{s}$.

b. $d_{\text{trans}} = \frac{L}{R}$.

c. $d_{\text{e2e}} = d_{\text{prop}} + d_{\text{trans}} = \frac{m}{s} + \frac{L}{R}$.

d. At time $t = d_{\text{trans}}$, the transmission process has just finished and Host A has just pushed the last bit onto the link, at the very beginning.

e. Since d_{prop} is greater than d_{trans}, the first bit would still be traveling in the middle of the road at $t = d_{\text{trans}}$ yet to reach Host B.

f. Since d_{trans} is greater than d_{prop}, the first bit would have already reached Host B at $t = d_{\text{prop}}$, which means when $t = d_{\text{trans}}$, the first bit is at Host B.

g. In this case, there are:

$$d_{\text{prop}} = \frac{m}{s} = \frac{m}{2.5 \times 10^8}, \tag{1}$$

$$d_{\text{trans}} = \frac{L}{R} = \frac{120\text{bits}}{5.6 \times 10^4\text{bps}}. \tag{2}$$

Let $d_{\text{prop}} = d_{\text{trans}}$, there is $m = \frac{120}{5.6\times10^4} \cdot 2.5 \times 10^8 \approx 5.36 \times 10^5$.

P31

a. $t_{\text{e2e}} = 3 \times \frac{8\times10^6 \text{ bits}}{2\times10^6 \text{ bps}} = 12$ s.

b. When the first packet arrives at the first switch, there is:

$$t_1 = \frac{10^4 \text{ bits}}{2 \times 10^6 \text{ bps}} = 5 \times 10^{-3} \text{ s.}$$

The time it takes for the second packet to arrive at the first switch is:

$$t_2 = \frac{10^4 \text{ bits}}{2 \times 10^6 \text{ bps}} = 5 \times 10^{-3} \text{ s.}$$

So after $t_1 + t_2 = 0.01$ s, the second packet arrives at the first switch.

c. $t = 3 \times \frac{10^4 \text{ bits}}{2\times10^6 \text{ bps}} + 799 \times \frac{10^4 \text{ bits}}{2\times10^6 \text{ bps}} = 802 \times 0.5 \times 0.01 = 4.01$ s.

d. Using message segmentation can:

1. **Increase the use rate of all the links**: in the context of this two-packet switch setting, all three links can be in use simultaneously most of the time.
2. **Reduce the cost in cases of error**: if an error occurs, only one small portion of the file (i.e., the specific erroneous packet) requires re-send instead of the whole big file.
3. **Reduce switches' waiting time**: the content stored by switches will be smaller.

e. Using message segmentation requires **additional header information bits** for all packets, introduces **processing cost because** of the segmentation itself, and **risks chances of missing packets or mis-ordered packets**.

2 Chapter 2

P17

a. POP3 transaction in the download-and-delete mode.

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: blah blah ...
S: .........blah
S: .
C: dele 1
S: +OK message 1 deleted
C: quit
S: +OK POP3 server signing off (deleting pending messags)
```

b. POP3 transaction in the download-and-keep mode.

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: blah blah ...
S: .........blah
S: .
C: quit
S: +OK POP3 server signing off
```

c. POP3 transaction in the download-and-keep mode describing the second POP session.

```
C: list
S: 1 498
S: 2 912
S: .
C: retr 1
S: blah blah ...
S: .........blah
S: .
C: retr 2
S: blah blah ...
S: .........blah
S: .
C: quit
S: +OK POP3 server signing off
```

Question.

a. Zoom uses the following 3 application-layer protocols, they are:

1. **HTTPS: used by all Zoom client applications and web browsers to transfer user content security**. ***Quote***: *All (Zoom) client applications and web browsers transferring user content must do so over end-to-end using TLS at every point of transfer. TLS (Transport Layer Security) protections enable browsers to transfer data through HTTPS, which enables authentication of the requested data* [1].

2. **WebSockets: used to ensure real-time bidirectional communication by offering a full-duplex communication channel over a single TCP connection.** ***Quote***: *Zoom WebSocket events all Developers to make a single, secured HTTP request to Zoom while keeping a single connection open to transmit all event data to the client*

[1] https://developers.zoom.us/docs/zoom-apps/security/#transport-layer-security-tls

subscribing to the event. Zoom delivers events when the WebSocket connection is open. When the connection is closed, Zoom does not deliver events. Any events that occur when the connection is closed will not be delivered. Once the connection is open again, Zoom will deliver events that occurred after the connection was reopened [2].

3. **SRTP: used to secure file transfer for media encryption and storage**. ***Quote****: The Zoom desktop client, mobile app, and web browser/client connections encrypt call media to the Zoom Cloud using SRTP with AES 256-bit encryption algorithm. SIP devices configured with SRTP use AES-128 or AES-256 bit algorithm to encrypt call media for connections to the Zoom Cloud, otherwise, unencrypted RTP is used as a fallback* [3].

b. Zoom is primarily based on a client-server architecture because meetings are hosted and routed through Zoom's globally distributed cloud servers and multimedia routers. Although some processing is performed on client devices to optimize performance, the core communication, routing, and scalability depend on Zoom's centralized infrastructure rather than direct peer-to-peer connections[4].

c. Zoom primarily uses the UDP at the transport layer for transmitting real-time media [5]. This choice is in alignment with the requirements for zoom's core business, i.e., real-time video, audio, and screen sharing, etc, because UDP offers lower latency, less overhead, and better performance under variable network conditions, which are essential for real-time communications.

d. No. A student's computer does not obtain the lecturer's IP address because Zoom uses a cloud-based client–server model. Both users send and receive media through Zoom's servers, not through direct peer connections.

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[2] `https://developers.zoom.us/docs/api/websockets/`

[3] `https://support.zoom.com/hc/en/article?id=zm_kb&sysparm_article=KB0069186`

[4] `https://www.sjsu.edu/workanywhere/docs/Zoom%20Message.pdf`

[5] `https://media.zoom.com/download/assets/Zoom+Connection+Process+Jan-09-2025.pdf/449a2ed2f05711ef907d5e9c08889a4d`